

# Investigating Psychological Value-Based Decision Making in Virtual Reality

Ellie Ebrahimi\*  
Department of Computer  
Science  
UNC Wilmington

Dale Cohen†  
Department of Psychology  
UNC Wilmington

Richard Pond‡  
Department of Psychology  
UNC Wilmington

## ABSTRACT

Psychological value-based decision making has been widely studied in traditional laboratory and real-world contexts, yet these approaches often lack ecological variability and immersive engagement. Virtual Reality (VR) offers a unique opportunity to simulate complex, dynamic environments that can reveal underlying mechanisms of value-driven choices. This ongoing project aims to examine how psychological values, such as perceived trustworthiness, intelligence, and physical attributes, influence decision-making processes, including choice and reaction time, in immersive VR scenarios. Participants will complete a series of forced-choice tasks embedded in realistic virtual contexts, with eye-tracking data collected to assess attentional patterns. By leveraging VR's flexibility, this study seeks to expand current models of value-based decision-making and to provide insights applicable to domains such as hiring, consumer behavior, and moral judgment.

**Index Terms:** Virtual Reality, Value-Based Decision Making

## 1 INTRODUCTION

Everyday decisions, from mundane selections to consequential judgments, are guided by psychological values attached to options and attributes, shaping both which choice is made and how quickly it is made. Classical laboratory work has established that value-based decisions can be captured by evidence accumulation models (e.g., drift diffusion), linking choice and response time to latent cognitive parameters such as drift rate, decision threshold, and non-decision time. These models have successfully explained preferential choices under time pressure, extended to contexts beyond perception (food choice, moral decisions, altruism) [5].

Beyond accumulation dynamics, context-dependent values play a crucial role. Recent work shows value representations are normalized relative to alternatives (e.g., via range normalization), altering choice consistency and reaction times as value ranges or option sets change. Meanwhile, reinforcement learning (RL) accounts capture how experience updates value estimates over time and can be integrated with diffusion models (RLDDM), bridging learning and choice dynamics [1, 3].

However, traditional studies often rely on static 2D stimuli and text-only scenarios, which can attenuate emotional engagement and limit ecological validity. Immersive VR provides presence and dynamic control over multimodal cues (space, time, interaction), enabling researchers to test whether decision processes observed in 2D tasks generalize to realistic, embodied settings. Systematic reviews across domains (sports decision training, neuropsychological assessment) highlight VR's promise for ecological validity and transfer, while also noting the need to measure attentional

and motor dynamics within immersive tasks. In moral decision-making, VR studies reveal divergences between judgments in text and actions in immersive scenarios, underscoring the methodological value of VR for probing affect, context, and embodiment [4, 2, 8, 6].

A critical advantage of modern VR head-mounted displays is integrated eye tracking, enabling measurement of gaze allocation, fixation dynamics, and attentional bias during decision formation. Cross-device evaluations show high central field accuracy with caution in peripheral regions, supporting research designs that place key stimuli centrally and log gaze dynamics as predictors of choice. Eye tracking in VR has already replicated the gaze cascade model in product choice, demonstrating that pre-decision gaze patterns predict subsequent selections, providing a window into online valuation processes. Together, these developments motivate a comprehensive VR framework to study psychological value-based decision making with stronger ecological validity and richer mechanistic reasoning [7].

## 2 METHOD

**Participants & Screening.** Adult participants (18+) will be recruited via campus platforms and community outreach. Exclusion criteria include severe motion sickness, uncontrolled migraines, or vestibular disorders. Demographic data (age, sex assigned at birth, gender orientation, ethnicity/race) will be collected via secure surveys; all non-age items include a "prefer not to say" option to minimize identifiability.

**Apparatus.** The experiment runs on a wireless Vive Pro Eye HMD with integrated eye tracking, calibrated per session. We adopt the best practice placement of decision-critical stimuli in the central visual field to maximize gaze accuracy and precision. Participants will interact with the simulation using controllers for selection. In the Desktop condition, participants will experience a simulation similar to VR with eye-tracking.

**Design & Tasks.** Participants will wear a wireless VR headset with built-in eye tracking and use handheld controllers to interact with the virtual environment. Before starting the main task, they will spend about 5 minutes learning to navigate and respond in VR and calibrating the eye-tracker. The main part of the study lasts about 30 minutes and consists of a series of decision-making trials. In each trial, participants will choose between two virtual avatars based on information about their traits, such as "trustworthy" or "intelligent." This is similar to the Harvard IAT in terms of its mechanics. The experiment includes two presentation conditions to create variation (Figure 1):

- In the Trait First condition, participants see the trait descriptions for both avatars before the avatars appear.
- In the Avatar First condition, participants see the avatars first, followed by the trait descriptions.

After all information is available, participants make a forced-choice decision by selecting one of the two avatars using the VR controllers. The system records their choice, reaction time, and

\*e-mail: ebrahimie@uncw.edu

†e-mail: cohend@uncw.edu

‡e-mail: pondr@uncw.edu

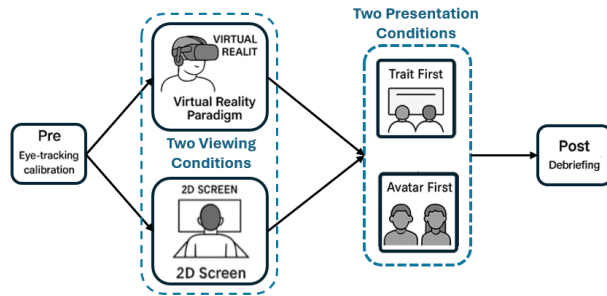


Figure 1: Experiment Design. Participants will be assigned into one of the viewing conditions where they experience the value-based decision-making tasks either in VR or a 2D Screen. There will be presented with two different presentation conditions; Trait First or Avatar First.

eye-tracking data (including gaze patterns and fixation times). All tasks occur in a safe, confined area, and participants remain seated as instructed. At the end of the study, participants complete a short demographic survey and receive credit for participation.

For the 2D screen condition, participants complete the same decision-making tasks on a standard computer monitor instead of in VR. They will use a mouse or keyboard to make selections, and the same two conditions (Trait First and Avatar First) apply. The timing and presentation of information mirror the VR setup, ensuring consistency across modalities. Eye-tracking calibration is performed before both conditions, and gaze data is collected using the integrated tracking system.

### 3 EXPECTED RESULTS

We expect that participants will make faster and more confident choices when one of the virtual individuals has traits that are considered highly valuable, such as being “intelligent”, or “trustworthy.” In other words, the more positive the attribute, the quicker and more consistent the decision. We also anticipate that the overall pattern of choices will depend on the context, such as the number of options shown and how the traits compare with one another, because people tend to judge values relative to what else is available. When participants receive feedback during the task, we expect their decisions to become quicker over time as they learn from experience. Eye-tracking data should reveal that participants look longer at the person they eventually choose and that their gaze shifts toward high-value traits earlier in the decision process. Finally, we believe that using VR will make these effects stronger and more realistic than traditional computer-based tasks because the immersive environment feels more like real-world decision-making.

### 4 DISCUSSION

**Understanding the theory.** Using VR gives us the ability to change many aspects of a situation at once, such as where participants stand, when information appears, and what social cues are present, while also tracking how participants look at things and make decisions. If we find that VR makes these patterns stronger or more consistent, it would support the idea that decision-making is not just a step-by-step process but a dynamic interaction between what we see, what we value, and how we act. Seeing participants adjust their choices based on the range of options in VR would also confirm that context matters and show how these effects work in more realistic, embodied settings.

**Improving research methods.** Our setup combines eye tracking with decision models to understand not just what participants choose, but how they arrive at that choice. By carefully placing im-

portant information in the center of the view and controlling timing, we ensure the data we collect is accurate and reliable.

**Practical applications.** The scenarios we use, such as hiring decisions, shopping choices, or moral dilemmas, are similar to real-world situations. This means we can study how participants handle bias, time pressure, and framing in ways that matter outside the lab. Past research shows that participants often behave differently in VR compared to text-based tasks, especially in moral decisions. Our platform leverages this to explore how emotions and physical presence influence choices.

**Managing challenges.** We will take steps to reduce discomfort, such as motion sickness or fatigue, by limiting movement, providing training, and monitoring participants throughout the study. We will also compare VR results with traditional computer-based tasks to identify the additional benefits VR provides and how well the findings might apply to other settings.

### 5 CONCLUSION

This project establishes a rigorous, mechanism-focused VR framework for psychological value-based decision making, integrating sequential sampling, value normalization, and gaze analytics in ecologically rich scenarios. By capturing how attention and context shape valuation and choice under immersion, the work advances theory and offers practical tools for assessment and training in domains such as hiring, consumer UX, and ethics education. Additionally, it positions VR as a high-validity testbed for decision models, with generalizable design guidance (centered on the participant’s visual field, timing manipulations, and calibrated eye tracking) and open datasets for reproducibility.

### REFERENCES

- [1] S. Bavard and S. Palminteri. The functional form of value normalization in human reinforcement learning. *Elife*, 12:e83891, 2023. 1
- [2] J. Connolly, D. Alder, M. Frame, and A. D. Wilson. Training decision making in sports using virtual reality: a scoping review. *International Review of Sport and Exercise Psychology*, pp. 1–18, 2025. 1
- [3] L. Fontanesi, S. Gluth, M. S. Spektor, and J. Rieskamp. A reinforcement learning diffusion decision model for value-based decisions. *Psychonomic bulletin & review*, 26(4):1099–1121, 2019. 1
- [4] Y. Jia, X. Zhou, J. Yang, and Q. Fu. Animated vr and 360-degree vr to assess and train team sports decision-making: a scoping review. *Frontiers in psychology*, 15:1410132, 2024. 1
- [5] M. Milosavljevic, J. Malmaud, A. Huth, C. Koch, and A. Rangel. The drift diffusion model can account for the accuracy and reaction time of value-based choices under high and low time pressure. *Judgment and Decision making*, 5(6):437–449, 2010. 1
- [6] E. Niforatos, A. Palma, R. Gluszny, A. Vourvopoulos, and F. Liarokapis. Would you do it?: Enacting moral dilemmas in virtual reality for understanding ethical decision-making. In *Proceedings of the 2020 CHI conference on human factors in computing systems*, pp. 1–12, 2020. 1
- [7] A. Palacios-Ibáñez, J. Marín-Morales, M. Contero, and M. Alcañiz. Predicting decision-making in virtual environments: an eye movement analysis with household products. *Applied Sciences*, 13(12):7124, 2023. 1
- [8] M. Pucci and L. Gagliardi. Moral dilemmas in virtual reality: a narrative review of findings and methodological challenges. *Frontiers in Psychology*, 6:1–21, 2025. 1